

Sienna WAV frame — MANUFACTURING PACKAGE — index (rev A, 2026-10-03)

One folder per part: <PART>_drawing.pdf + DXF files. Units: inch. Quantities are per vehicle (FR = right side, FL = left side mirror).

Item	Description	Material	Flat / length (in)	Bends	Qty/veh	Process
FR-1	Plate with ears — flat pattern (2 bends 90°)	10 ga 0.135"	8.512 × 13.851	2	2	laser + brake
FR-2	FR-2 one piece (top + 2 U legs)	10 ga 0.135"	22.078 × 3.250	6	2	laser + brake
FR-3	Channel — flat pattern (2 bends 90°)	11 ga 0.120"	4.084 × 7.734	2	2	laser + brake
FR-4-1	Wrap strip (ear, top, front, bottom, slant)	10 ga 0.135"	18.746 × 2.716	9	2	laser + brake
FR-4-2	Side plate A (2 holes Ø0.512)	10 ga 0.135"	6.500 × 6.875	0	2	laser
FR-4-3	Side plate B (holes Ø0.512 + Ø0.34)	10 ga 0.135"	6.500 × 6.875	0	2	laser
FR-4 tube	Tube OD 1.5 / ID 1.181 — cut 3.689" (through bottom and to	DOM tube OD 1.500" × ID 1.181"	L 3.689	0	2	saw
FR-5	U-body (2 sides + plate A-C)	10 ga 0.135"	10.643 × 6.198	2	2	laser + brake
FR-6 tube	Tube 3×3 — cut 14 ¼"	Sq. tube 3" × 3" × 3/16" wall	L 14.250	0	2	saw
FR-7	FR-7 one piece (back + bottom + 2 sides)	10 ga 0.135"	9.801 × 8.160	3	2	laser + brake
FR-8	Channel (top + 2 sides)	11 ga 0.120"	6.209 × 12.000	2	2	laser + brake
FR-9 tube	Tube 3×3 — cut 8"	Sq. tube 3" × 3" × 3/16" wall	L 8.000	0	2	saw
FR-10-1	Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends	10 ga 0.135"	12.287 × 11.214	3	2	laser + brake
FR-10-2	Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)	10 ga 0.135"	12.478 × 2.716	3	2	laser + brake
FR-10 tube	Tube OD 1.5 / ID 1.181 — cut 3.42" (through A-B and the sh	DOM tube OD 1.500" × ID 1.181"	L 3.420	0	2	saw
FR-11	FR-11 plate (flat)	11 ga 0.120"	5.125 × 2.000	0	4	laser
FF-1	Channel with ears — flat pattern, one piece (4 bends 90°,	10 ga 0.135"	20.139 × 6.818	4	1	laser + brake
FF-3-1	FF3-1 top plate — flat pattern, 10 bends 90° down	11 ga 0.120"	39.403 × 15.073	10	1	laser + brake
FF-3-2	FF3-2 lower plate — flat pattern, 2 bends 90° up	11 ga 0.120"	39.000 × 14.595	2	1	laser + brake
FF-3-3	Passage wall 1⅝ × 4.69" (instead of the tube)	11 ga 0.120"	4.687 × 1.125	0	2	laser
FF-3-4	Passage wall 1⅝ × 5.01" (instead of the tube)	11 ga 0.120"	5.013 × 1.125	0	2	laser
FF-4 tube	Square tube 1 ½" — cut 39"	Sq. tube 1 ½" × 1 ½" × 3/16" wall	L 39.000	0	1	saw

MATERIAL PER VEHICLE (net, without nesting scrap)

- 10 ga 0.135" (3.42 mm): 907 in² net (6.30 ft²), ≈ 34.6 lb
- 11 ga 0.120" (3.04 mm): 1214 in² net (8.43 ft²), ≈ 41.2 lb
- DOM round steel tube OD 1.500" × ID 1.181" (wall ≈ 0.160"): 14.22 in total length
- Square steel tube 3" × 3" × 3/16" wall (A500): 44.50 in total length
- Square steel tube 1 ½" × 1 ½" × 3/16" wall (A500): 39.00 in total length